
How Separation Weight Governs Flock Cohesion in a Vicsek-Inspired Boid Model

Group 19

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Abstract

Flocking is a key example of collective intelligence, where simple local interaction rules can generate global coordinated motion without a central method of control. The standard boid model codes motion via cohesion, alignment, and separation – the three local forces. The separation force is commonly kept constant rather than manipulated as a variable. This study asks how the separation weight affects the cohesion of the flock, measured by the steady-state average nearest-neighbor distance (ANND), in a two-dimensional Vicsek-inspired boid model. Varying the separation weight across five values with ten matched random seeds (50 runs), it was found that ANND increases monotonically and gradually with the separation weight; there is a 34% increase in the range of values. The effect is largely significant (repeated-measures ANOVA) with $F(4,36) = 39.86$, $p < 0.001$, $\eta_p^2 = 0.82$. In addition, it was confirmed by a Friedman test. The relationship is gradual for all steps except the two smallest adjacent ones, which are statistically indistinguishable. Repulsion is strengthened with no abrupt loss of cohesion in the range studied.

1 Introduction

Collective intelligence is the study of how agents adhering to simple rules with no central control or global plan, give rise to group behavior that is coordinated [4, 5]. A fundamental question for such a collective system is what maintains said coordinated behavior of the group, and the nature of this property's degradation as the interactions between the individuals in the group change. Predictable degradation implies control over the degradation as there is a degree of understanding coming from predictability. However, a system that degrades abruptly (at some unknown value) cannot be controlled. Such a distinction is important for studying and explaining natural patterns of collective intelligence such as bird flocks, fish schools, artificial ones like swarm robotics.

Flocking is among the clearest examples of this phenomenon. In the boid model, each agent is 'directed' with three local forces: alignment, cohesion, and separation [4]. Alignment denotes the agent matching the heading of nearby agents, cohesion, the agent moving towards their average position, and separation, moving away from the agents that are too close. The balance between these three forces dictates whether the group maintains cohesion, and manipulating it can alter the collective state of the group [2].

Separation is the only purely repulsive force, and while noise and density have been extensively studied in self-propelled particle models, separation is generally held fixed rather than being treated as a variable [5, 3]. We therefore isolate separation and how its strength governs cohesion is put into question.

Concretely: *how does the separation weight affect flock cohesion, measured as the steady-state average nearest-neighbor distance (ANND), in a Vicsek-inspired boid model?* Answering the research question above does not only uncover whether stronger repulsion loosens the flock, but whether it is done gradually or abruptly; with an instant loss of cohesion. This distinction determines how predictable the nature of the collective behavior is.

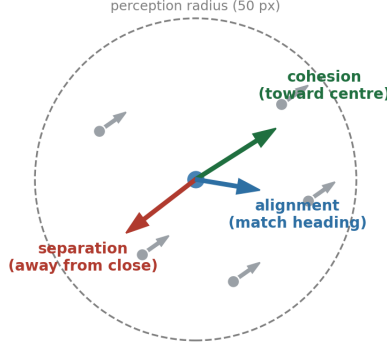


Figure 1: One agent’s decision. From neighbors within its perception radius it computes the weighted sum (Eq. 1) of alignment, cohesion, and separation steering it while its speed stays constant.

It is hypothesized that increasing the separation weight increases the ANND *monotonically and gradually*. The underlying mechanism is that strengthening the separation force shifts the balance between the other forces smoothly rather than switching between discrete states, since the forces are a weighted sum. This prediction is directional and falsifiable. The alternative that cohesion collapses abruptly past a critical value was also considered; similarly Couzin et al. reported sharp behavioral transitions and a sudden jump in ANND when zone geometry changes [2]. The rest of the report tests the two predictions using a controlled sweep and a within-subjects statistical analysis- finding support for the gradual hypothesis.

2 Methodology

Background: Our report uses three important parts of flocking literature. Vicsek et al. [5] introduced self-propelled particles that move at constant speed and align with neighbors, with noise, and summarized the collective state in a single scalar order parameter – the idea the report adopts but applied to a spatial metric. Couzin et al. [2] modelled three concentric interaction zones (repulsion, orientation, attraction) that map directly onto separation, alignment, and cohesion, and showed that small changes in zone geometry produce sharp transitions between collective states, motivating the need to isolate a single force. Ballerini et al. [1] found that real bird flocks interact topologically; with a fixed number of nearest neighbors. Contrary to interacting within a fixed metric distance, which informs our discussion of limitations.

Model and agent design: We implement Reynolds’ three-rule boid model [4] with Vicsek-style constant-speed dynamics, in two dimensions and without an explicit noise term, using the Violet (vi) framework. Each agent has a state (a position and a velocity). Its perception is local: it responds only to neighbors within a fixed radius $r = 50$ px. Its decision combines three force modules computed from those neighbors (Fig. 1):

- **Alignment:** the average neighbor velocity minus the agent’s own velocity.
- **Cohesion:** the vector from the agent toward the average neighbor position.
- **Separation:** the average of unit vectors pointing away from each neighbor.

The modules combine into a single steering force, scaled by mass m ,

$$\mathbf{f}_{\text{total}} = \frac{w_a \mathbf{a} + w_c \mathbf{c} + w_s \mathbf{s}}{m}, \quad (1)$$

where w_a, w_c, w_s are the alignment, cohesion, and separation weights. The *action* is to add $\mathbf{f}_{\text{total}}$ to the velocity, cap the speed at a constant maximum (the Vicsek-style constraint, so only direction changes), and update position. The outcome metric is the ANND: for each agent we take the distance to its single nearest neighbor, average over all agents, and average that over the steady-state window. A low ANND means a tight, cohesive flock; a high ANND means a dispersed one.



(a) $w_s = 0.1$: tight, cohesive flock.



(b) $w_s = 1.0$: dispersed flock.

Figure 2: Representative steady-state configurations of the 50 agents at the lowest and highest separation weights.

3 Experiments

3.1 Experimental setup

The separation weight was varied across five conditions: 0.1, 0.3, 0.5, 0.7, and 1.0. Ten random seeds were run per condition, giving 50 simulation runs. The same ten seeds were reused in every condition, so each seed fixes one initial configuration that is then observed under all five weights. All other parameters were held fixed (Table 1). Each run lasted 500 ticks; the first 200 were discarded as a stabilization period, and the ANND was averaged over ticks 200–500 to give one value per run.

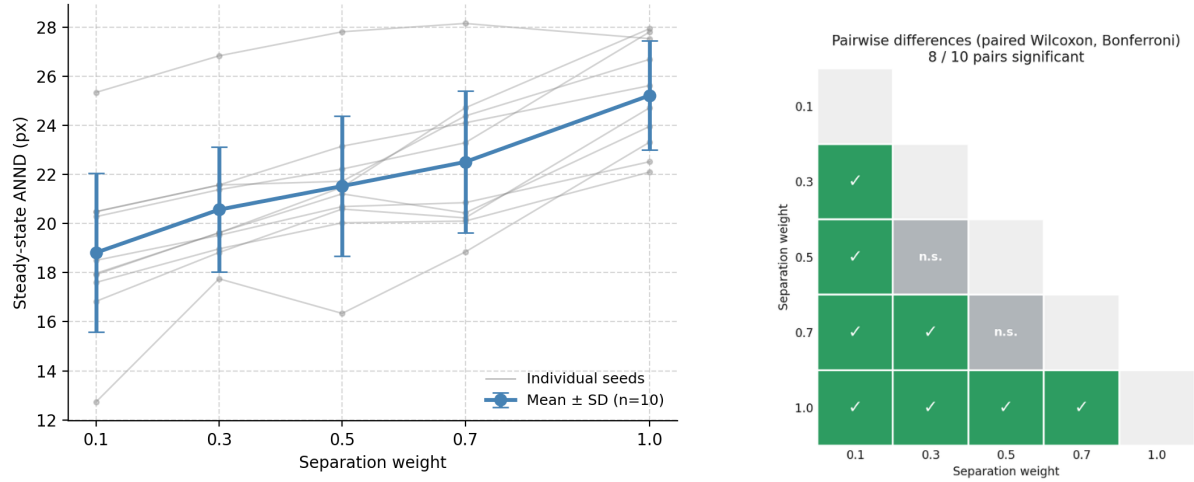
Table 1: Fixed and manipulated parameters used in all experiments.

Parameter	Value	Status
Number of agents	50	Fixed
Alignment weight	0.5	Fixed
Cohesion weight	0.1	Fixed
Perception radius	50 px	Fixed
Max velocity	$\sqrt{2} \approx 1.41$	Fixed
Ticks per run	500	Fixed
Stabilisation period	200 ticks	Fixed
Seeds per condition	10	Fixed
Separation weight	0.1, 0.3, 0.5, 0.7, 1.0	Manipulated

Because the same seeds appear in every condition, the design is within-subjects: each seed is a matched configuration measured under all five weights, not an independent sample. We confirmed this directly – ANND values correlate strongly between conditions across seeds (Pearson $r \approx 0.6$ – 0.97), far from the near-zero expected if conditions were independent. We therefore assess the effect with a *repeated-measures* ANOVA, appropriate for a continuous outcome across five ordered conditions sharing matched seeds, which removes between-seed variance from the error term rather than leaving it there. As the residuals were mildly non-normal (Shapiro–Wilk $p = 0.044$) while variances were homogeneous (Levene $p = 0.88$), we cross-validated with the non-parametric Friedman test. Post-hoc comparisons used paired Wilcoxon tests with Bonferroni correction, and we fitted a linear regression of ANND on the separation weight to capture the trend.

3.2 Results

Steady-state ANND rises monotonically with the separation weight, from 18.8 px (95% CI [16.5, 21.1]) at $w_s = 0.1$ to 25.2 px ([23.6, 26.8]) at $w_s = 1.0$ – a rise of about 34% (Fig. 3a). The effect is large and highly significant: the repeated-measures ANOVA gives $F(4, 36) = 39.86$, $p < 0.001$, with partial $\eta_p^2 = 0.82$, and the Friedman test agrees



(a) ANND vs. separation weight. Bold line = mean $\pm$ SD across 10 seeds; faint lines = individual seeds.

(b) Pairwise comparisons (paired Wilcoxon, Bonferroni). Green = significant, grey = n.s.

Figure 3: Effect of separation weight on steady-state ANND across 50 runs.

($\chi^2(4) = 35.6$, $p < 0.001$; Kendall's $W = 0.89$). A linear fit gives a slope of 6.76 px per unit of separation weight ($R^2 = 0.39$, $p < 0.001$); the modest R^2 reflects that most remaining variance is between seeds, which the within-subjects analysis isolates. For comparison, an independent one-way ANOVA on the same data yields only $F = 7.40$ — much weaker precisely because it leaves the large between-seed variance in the error term, which also explains why the per-condition standard deviations look so similar (the same seeds drive the spread in every condition).

The pairwise tests locate the effect (Fig. 3b): 8 of 10 comparisons are significant after correction. The weakest and strongest conditions (0.1 and 1.0) each differ from every other condition; the only non-significant pairs are 0.3–0.5 and 0.5–0.7, the two smallest adjacent steps.

Viewing all 50 runs at once (Fig. 4) makes both patterns visible in one picture: every row (seed) warms from left to right, showing the effect is consistent across seeds rather than driven by a few; and some rows are persistently warm while others are persistently cool, which is the matched between-seed structure the repeated-measures analysis accounts for.

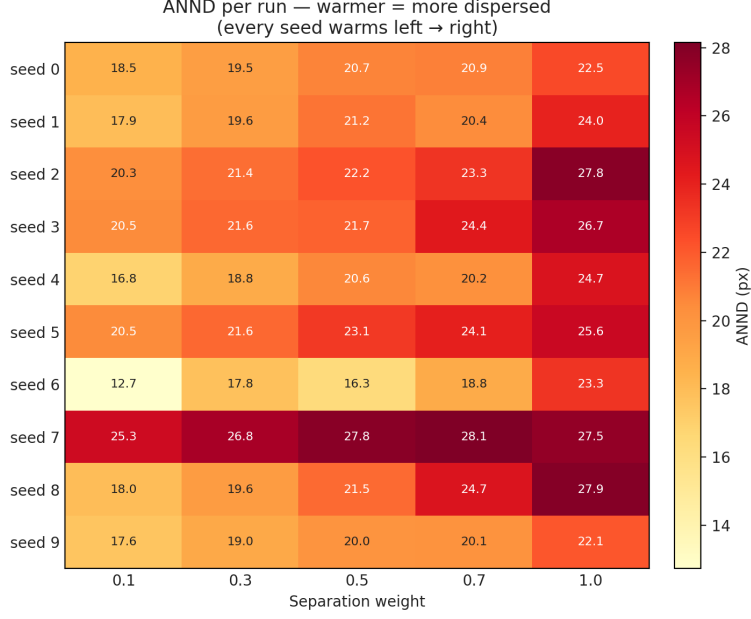


Figure 4: Steady-state ANND for every run (10 seeds $\times$ 5 conditions). Warmer cells are more dispersed. Each row warms left to right (the effect); warm and cool rows persist across conditions (matched between-seed variance).

The verdict: the flocks do not split up abruptly due to a repulsion transition, but rather the separation effect increases smoothly. This claim is proved by pairwise tests since if a critical value existed, then there should have been a single pair of adjacent points that demonstrated a huge jump. However, it is seen that the most extreme points (the comparisons between 0.1 and 1.0) are significant relative to all other values, while the two pairs that are not significant at all happen to be located in the middle of the graph. This pattern is consistent with a gradual and monotonic effect of the separation weight on ANND, supporting our hypothesis.

4 Conclusion

Our findings have indicated that as the separation weight increases, the cohesiveness of the flock reduces gradually, not abruptly. The gradual spread of the agents is attributable to the weighted sum of the forces; an increase in the separation weight causes a steady shift in the ratio of the forces. The property of the separation force 'self-limiting' itself, as well as the relationship between the distance between the agents and the separation force, makes ANND rise steadily and not infinitely. This is unlike the cases where repulsion takes strict precedence and leads to sharp behavioral transitions [2], but is in agreement with the continuous nature of the classic Vicsek transition [5]. More generally, our findings imply that the degradation or catastrophic collapse of a collective system is determined by the way the forces underlying it are combined together. In a Vicsek-inspired boid model, the separation weight has a continuous, statistically significant, and monotonic effect on flock cohesion.

There are several important limiting conditions in our study. In our study, the agents use and have a constant metric perception radius, whereas flocks have a topological one in reality, which may help real flocks maintain cohesion better at varying densities [1]. The three forces also differ by order of scale, hence the weights used are not directly comparable. Therefore, the bounded separation force may never fully dominate the distance-scaled cohesion force within the range we tested, and so the absence of an observed fragmentation point does not prove that no such point exists. Lastly, ANND measures only the spacing, not whether the coordinated movement is retained. We only ran our simulations with ten seeds and a single unit-vector separation rule.

Our future research can include polarisation measurement along with ANND in order to test whether the coordination survives when the flock disperses. Another area of research can be broadening the scope of the separation sweep in order to find the fragmentation threshold and mapping separation against cohesion forces on a two-dimensional grid. Lastly, the implementation of a topological rule for the perception radius would make our model more realistic. Each of these areas can be developed using the framework established in this paper.

References

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